Rassah (رصّة): AI Packing Assistant. Project Report

Project 1: AI Product Development and Monitoring · Tuwaiq Academy × Startupbootcamp × Meta · 22 October 2026 **Team:** M1 (AI & Data) · M2 (Engine & UI) · M3 (Platform & Monitoring) **Date:** 2026-10-22 **Live demo:** https://rassah.vercel.app · **Repository:** https://github.com/justAbdulaziz10/Rassah

> Every number comes from `eval/results/*.json`, `docs/report/results.md`, `docs/demo-script.md` or the dashboard.

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1. Problem

Merchants who ship multi-item orders choose cartons by eye. Three things go wrong:

1. **Oversized cartons raise shipping cost.** Couriers bill the larger of actual and volumetric weight (`L × W × H / 5000`).
2. **Fragile items break and liquids leak.** A new packer puts a book on a perfume bottle, or lays an oil bottle flat.
3. **The data that would fix this doesn't exist.** Professional cartonization tools assume clean product dimensions in a warehouse system. Small Saudi merchants only have free-text product pages in Arabic and English.

Validation status: merchant outreach: no stores contacted and no replies recorded yet (log with empty rows: `docs/discovery.md`). The problem size is a hypothesis we have not measured.

2. Solution and user flow

The packer types an order number. Rassah:

1. Loads the order's products. For each product not already cached, **Gemini reads the product page** and returns outer-package dimensions, weight, fragile / liquid / sensitive flags, where each value came from (*page*, *inferred*, *unknown*), confidence, and an evidence quote.
2. Runs a **constrained 3D packing search** over the carton catalog: gravity (≥ 70% base support), nothing on fragile items, liquids upright, weight limits, a safety margin on inferred sizes, and padding for fragile/sensitive items.
3. Gates every layout through an **independent validator**, then picks the cheapest valid carton by estimated carton + shipping cost.
4. Shows the carton, **numbered Arabic packing instructions**, a 3D step-by-step view, estimated cost, and per-product source badges. The packer **confirms or corrects**. Corrections are stored and measured.

If a value has no basis, the order stops with "needs measurement" (the packer measures once). If no carton fits within the search budget, it says so. It never invents a layout.

رصّة الطلبات الكتالوج المراقبة التقييم

تجهيز طلب

اكتب رقم الطلب، ورصّة تقرأ صفحات المنتجات وتقترح الكرتون الآمن الأوفر مع تعليمات التغليف.

رقم الطلب

1002

جهّز الطلب

محاكاة تعطل Gemini تحكم العرض

طلبات المتجر التجريبي

#1001 — 1 قطع عرض: نجاح سريع single page-sourced item, smallest carton	#1002 — 3 قطع عرض: قابل للكسر + سائل + تقديري inferred perfume (fragile + liquid) with a book and a charger	#1020 — 12 قطع عرض: مستحيل kg exceeds every carton's limit → no_box (overweight) 20.5»
#1003 — 2 قطع two liquids (honey + olive oil) must stay upright	#1004 — 3 قطع fragile + heavy mix: ceramic burner with dates and notebooks	#1005 — 3 قطع electronics bundle (sensitive padding)
#1006 — 2 قطع group C product → needs_measurement	#1007 — 3 قطع small flat items, should pick a small or flat carton	#1008 — 2 قطع porcelain mugs + glass candle (two fragile items)
#1009 — 4 قطع skincare set: plastic liquid + glass liquid (inferred)	#1010 — 5 قطع coffee, tea and incense (inferred, non-fragile)	#1011 — 3 قطع hair dryer + accessories
#1012 — 3 قطع empty glass bottle (fragile, not liquid) + books	#1013 — 2 قطع large porcelain plate set + dates	#1014 — 3 قطع three perfumes (page + inferred)
#1015 — 6 قطع stationery bulk order	#1016 — 4 قطع mixed gift basket: honey, dates, incense burner, coffee	#1017 — 2 قطع two group C products → needs_measurement

رصّة الطلبات الكتالوج المراقبة التقييم

← كل الطلبات

الطلب #1002

تم اختيار كرتون

كرتون كبير L 40 × 30 × 25 سم

3 من الكاش · 0 استُخرج بالـ AI · 232 ms

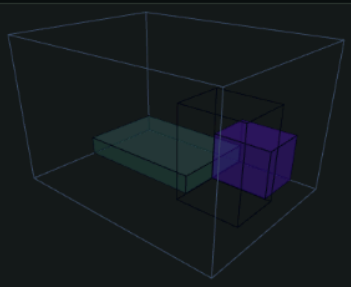
إعادة التجهيز

تأكيد التغليف

تعليمات التغليف

1. استخدم كرتون «كبير L» (40 × 30 × 25 سم من الداخل).
2. ضع «رواية ساق البامبو» في قاع الكرتون.
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
3. ضع «شاحن جداري سريع 20W» في قاع الكرتون.
   - حساس: أضف حشوة حماية 1.5 سم من كل الجهات.
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
4. ضع «عطر أو دو بارفان 50 مل» في قاع الكرتون.
   - قابل للكسر: لفّه بغلاف فقّاعي (1.5 سم على الأقل) ولا تضع فوقه أي شيء.
   - سائل: أبقِه واقفاً (هذا الجانب للأعلى).
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
5. املأ الفراغات بمواد حشو حتى لا تتحرك القطع، ثم أغلق الكرتون.
6. اكتب على الكرتون: «هذا الجانب للأعلى ↑».

العرض ثلاثي الأبعاد



الخطوة 3 / 3

عادي حساس سائل قابل للكسر الشفاف = مقاس تقديري

التكلفة التقديرية (أسعار تجريبية، ليست أسعار شركة شحن)

الكرتون	الوزن الفعلي	الوزن الحجمي	الشحن	الإجمالي
ر.س 2.60	كجم 1.25	كجم 6.61	ر.س 36.00	ر.س 38.60



3. Where the AI is, and why

Step	Technique	Why
Understand product pages	**LLM (Gemini)** with a JSON schema and honesty rules	Free-form Arabic/English text, mixed units, Arabic-Indic digits, implicit fragility ("كريستال") defeat rules
Arrange items	**Search** (classical AI): corner-grid candidates × rotations × 9 orderings/scores + 40 seeded restarts	An LLM cannot guarantee collision-free, supported coordinates
Prove validity	Independent validator (plain geometry)	Correctness checked by code that shares nothing with the search

"*The LLM understands, the engine computes, the validator proves.*" The instructor confirmed that using AI APIs meets the brief and that no fine-tuning is needed.

4. Data collection

- **Synthetic, stated openly.** 30 products written by the team in the style of Saudi store pages (no real merchant data):
  - **Group A (15):** dimensions and weight stated, in varied formats (Arabic-Indic digits, inches/lb, "الطول/العرض/الارتفاع", buried mid-paragraph).
  - **Group B (12):** clues only (50 ml, 400 pages, 12 pieces).
  - **Group C (3):** no usable information.
- **Ground truth** per product (outer box, natural upright orientation, flags, expected source). Independent review by a member other than the description author is still open (CHECKLIST P3.12),

so the ground truth is AI-drafted and unreviewed. **10-product holdout** is excluded from prompt tuning.
- **20 orders** (incl. 3 demo orders, needs-measurement and impossible cases), **5 cartons**, synthetic shipping rates.
- **60 packing fixtures** from a seeded generator: 20 known-feasible (guillotine cuts of a carton, so a witness layout exists), 20 impossible (oversize / overweight / liquid too tall), 20 random.
- All data files are schema-validated in CI (`npm run validate-data`).

5. Technical design

Mermaid sources are in `docs/diagrams.md`. PNG exports:

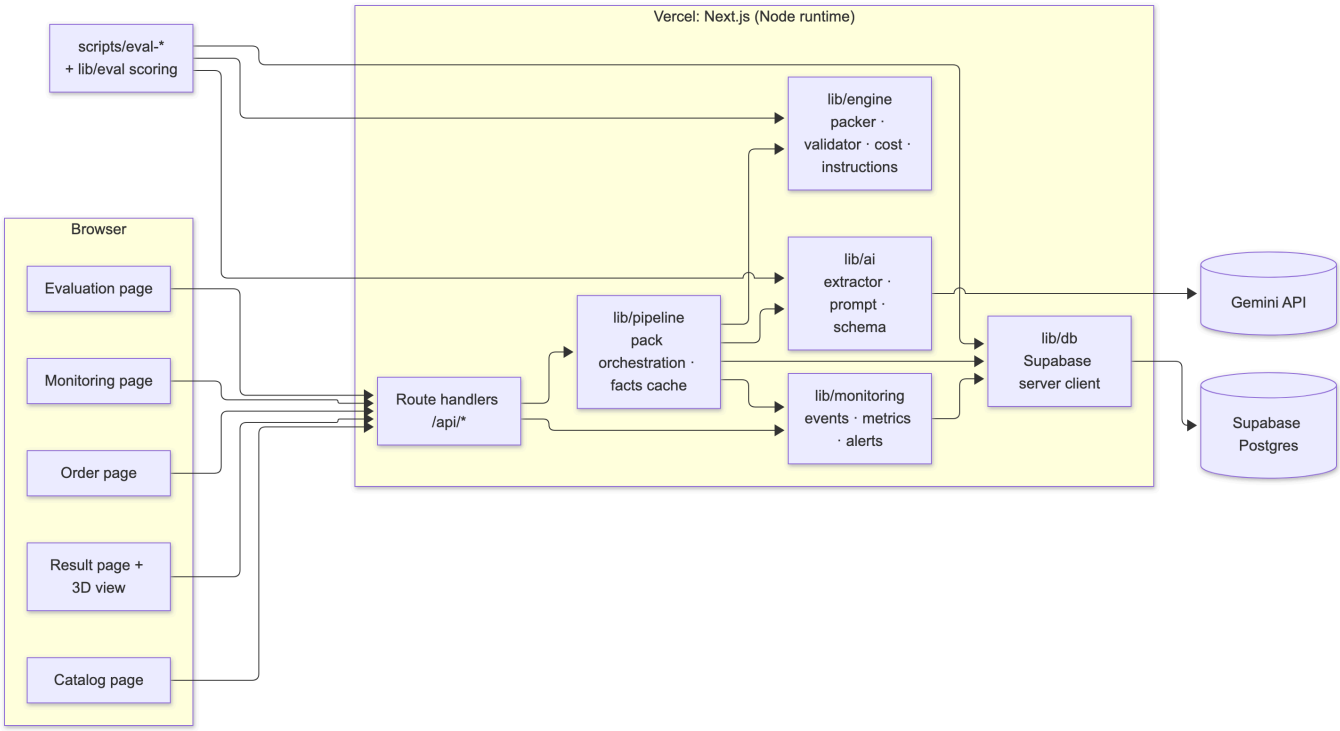


Figure 1. System architecture: pages → route handlers → pipeline → AI / engine / monitoring → Supabase.

- **Stack:** Next.js 16 (App Router, Cache Components) + TypeScript strict on Vercel. Supabase Postgres. `@google/genai`. react-three-fiber. Recharts. Vitest.
- **Pack pipeline** (`lib/pipeline/pack.ts`): order → catalog cache (source-hash freshness; manual/corrected facts never overwritten) → Gemini for misses (concurrency 2, 20 s timeout, retries only on 429/5xx, SDK retries off) → engine → validator → `order_runs` + `extraction_events` → alert refresh. The result is returned even if monitoring writes fail (flagged "monitoring degraded").
- **Engine** (`lib/engine/`): pure TypeScript, no I/O, deterministic. Every rule exists twice: in the search, and in the independent validator.
- **Cost model:** carton + shipping on chargeable weight. Illustrative rates, always labeled "estimated".
- **Security:** service-role key server-only, RLS on all tables with no public policies, Zod on every input, demo controls only with `DEMO_MODE=true`, product text delimited as data in the prompt.
- **Offline fallback:** `DATA_BACKEND=memory` runs the whole app without Supabase (local only).

6. Development environment

Branch → PR → **GitHub Actions CI** (lint, typecheck with `next typegen`, data validation, 152 tests) + **Vercel preview** per PR → teammate review → merge → production deploy → smoke test → dashboard. Decisions

are logged with reasons in `docs/decisions.md`. Progress is tracked in `docs/CHECKLIST.md` (517 tasks, 13 phases, gates). Exact dependency versions are pinned, and Node 24 LTS is set via `.nvmrc`.

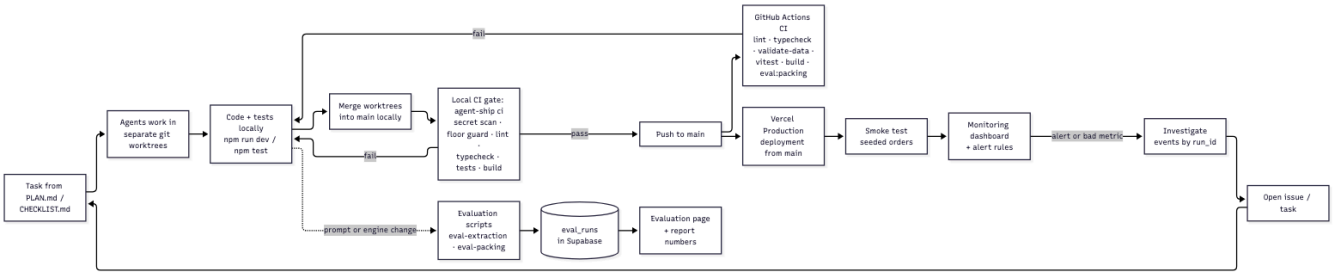


Figure 2. Development flow: branch → CI + Vercel preview → review → main → production → monitoring → issue.

7. Selected metrics (monitoring)

#	Metric	Reveals	Alert
M1	AI correction rate (reviewed AI products the packer corrected)	Wrong extractions	> 20% of last 30, warning
M2	Data-source mix of packed units	Missing store data	> 30% unknown in 24 h, info
M3	Validator rejections of engine layouts	Engine bugs (must be 0)	any in 24 h, **critical**
M4	No-carton rate + skip reasons	Catalog/margin mismatch	> 15% of last 20, warning
M5	p50/p95 latency, cached vs uncached	Slow Gemini vs slow engine/DB	p95 > 8 s uncached / 2 s cached
M6	Gemini failure rate by status	Key, quota, outage, bad output	> 10% of last 20, **critical**
M7	Cost per order + cache-hit rate	Cache broken, prompt bloat	(shown)

Every metric shows its sample size and says "insufficient data" below its minimum. Eval traffic is excluded. Alerts open once per rule, auto-resolve, and post to a webhook (never for simulated faults). Each run has a trace (Gemini calls, tokens, timings, corrections).

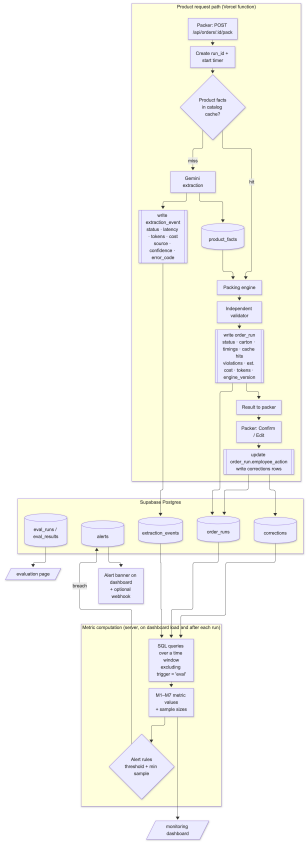


Figure 3. Monitoring flow: events written per run, metrics M1–M7 computed over a window, alert rules.

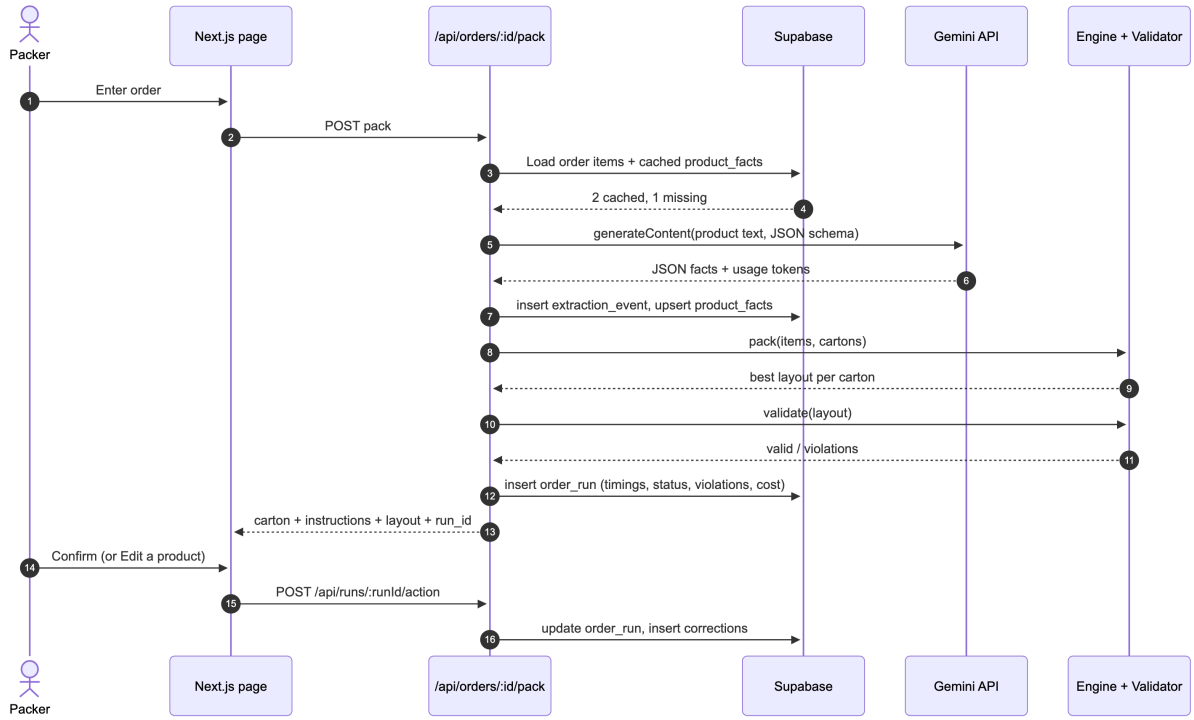


Figure 4. One order end to end (uncached product): what is written where.

8. Benchmarks and evaluation results

Packing (60 fixtures, `eval/results/packing-2026-10-21-engine-0.2.json`):

Metric	Result
Engine layouts passing the independent validator	**40 / 40 (100%)**
Known-feasible recall (zero-slack exact fills)	**20 / 20 (100%)**
False claims on impossible cases	**0 / 20**
Naive baseline layouts failing the validator	**20 / 28**: liquid tipped 17, on fragile 7, floating 2
Mean cost saving vs baseline (both valid, n = 8)	0 SAR (**no savings claimed**)
Runtime p50 / p95	0.3 / 12.9 ms

Evaluation-driven iteration: recall 3/20 (extreme points) → 9/20 (corner grid + seeded restarts; more restarts gave no gain) → 15/20 ("tight" placement scores, found by diagnosing fixture `feasible-05`) → **20/20** (engine-0.2: backtracking anchor search for the exact fills the greedy passes missed). The 0 SAR saving is by construction (both pick the cheapest fitting carton), not a bug.

Extraction (`eval/results/extraction-2026-10-21-v1.json`, `gemini-3.5-flash-lite`, prompt v1, 30 synthetic products, 30/30 calls ok). Table = run 2 (latest). Two runs were made (same model and prompt): estimated-dim MAPE 14.3% -> 12.9%, holdout 18.2% -> 16.1%, and one fragile false positive in run 2. This is LLM run-to-run variation; the committed file and the /evaluation page show run 2 (latest), run 1 is "previous".

Metric	All (30)	Tuning (20)	Holdout (10)
Parse success	100%	100%	100%
Dims within 1 cm / weight within 10% (group A)	100% / 100%	100% / 100%	100% / 100%
Inferred dim MAPE (group B)	12.9%	11.3%	16.1%
Inferred weight MAPE (group B)	13.5%	11.9%	16.7%
Source correctness	100%	100%	100%
Honesty on no-info products	100%	100%	100%
Fragile recall / precision	100% / 90.9%	100% / 87.5%	100% / 100%
Liquid accuracy	100%	100%	100%
Avg latency / tokens	1795 ms / 632	1976 ms / 632	1433 ms / 633
Cost per product	$0.000452	$0.000453	$0.000450

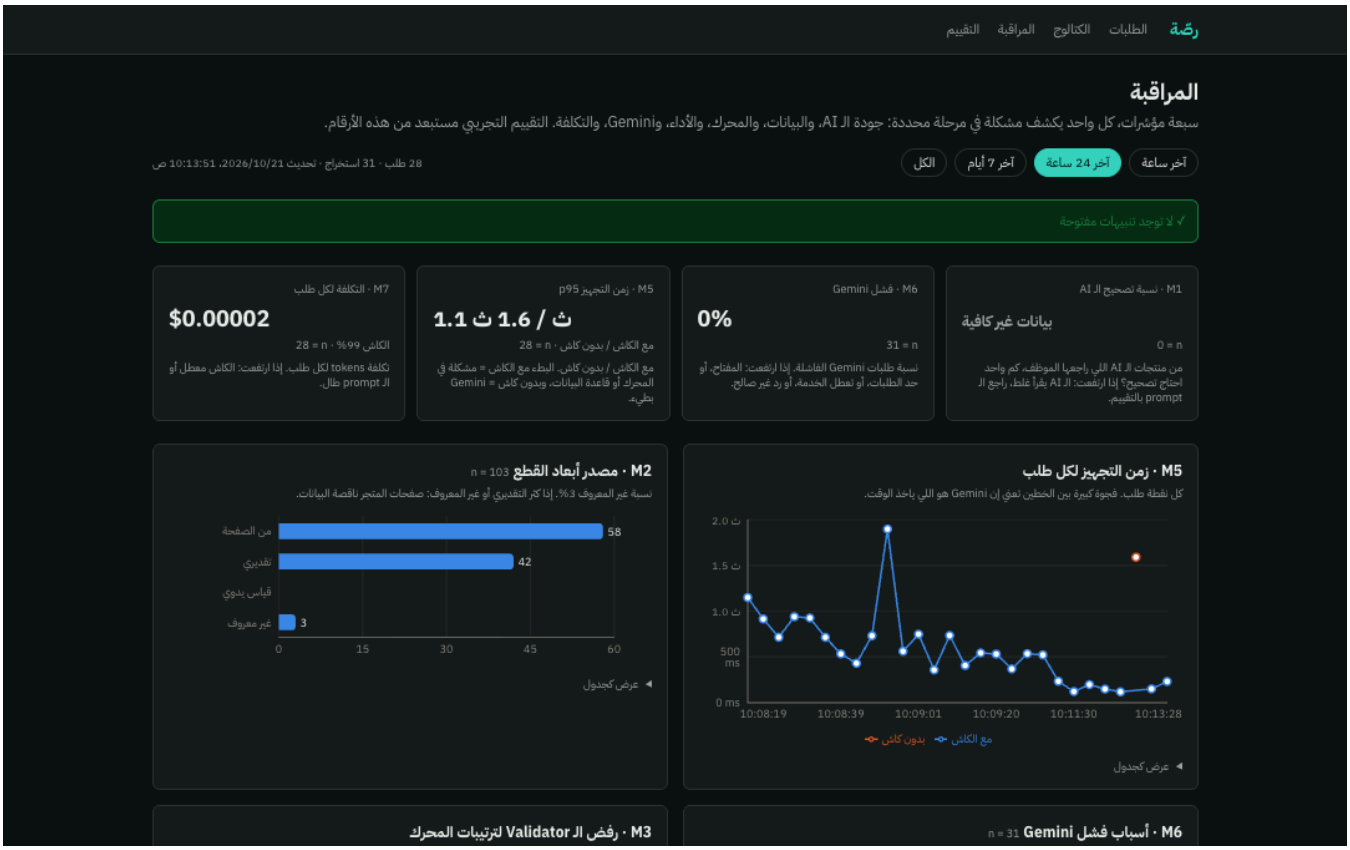
Prompt iteration: none. Tuning-set failures were limited to estimate error on inferred products (no extraction failures), so the prompt stays `v1` and there is no before/after row.

Margin decision: inferred dim MAPE over all 12 inferred products is 12.9%, rounded up to 15%, so `MARGIN_INFERRED` stays 0.15. The holdout MAPE (16.1%, 4 products) is above 15%. With so few inferred products this is noisy, and the instructions already say "check the fit". A real-merchant pilot should re-measure it before the margin is trusted or raised.

Model: `gemini-3.5-flash-lite` ($0.30 in / $2.50 out per 1M tokens, 2026-10-21). The faster-tier models had a free quota of 20 requests/day, too low for a 30-call eval.

9. Monitoring in action

- **Investigation 1: slow cached orders (2026-10-21).**
  - *Problem:* after the first production deploy, a smoke test of all 20 orders looked slow.
  - *Metric:* the M5 latency chart on `/monitoring` showed cached orders taking **358-1896 ms** server-side.
  - *Trace:* the `x-vercel-id` response header showed the functions running in `iad1` (US East), while Supabase is in `eu-central-1` (Frankfurt). Every DB query crossed the Atlantic.
  - *Fix:* `vercel.json` `"regions": ["fra1"]`, pushed the same day (logged in `docs/decisions.md`).
  - *After:* cached orders **119-234 ms**. One uncached order (Gemini call): **1589 ms** total, of which extraction 1205 ms. The drop is visible in the screenshot below.
- **Investigation 2: Gemini quota.** `rate_limited` events (429 QuotaFailure) showed the Flash models' free tier is 20 requests/day/model. We switched to `gemini-3.5-flash-lite` (500/day); see `docs/decisions.md`.
- **Investigation 3 (offline):** the packing eval and its fixture trace led to the scoring fix in section 8. A regression test guards it.
- **Live demo:** a simulated Gemini outage (demo toggle) turns uncached products into "needs measurement" while cached ones still pack. M6 crosses 10% and shows a critical banner "(عطل تجريبي)". The run trace shows `simulated_outage`. The alert resolves after clean runs.



10. Tools and AI usage

Summary of `docs/report/TOOLS.md`: Gemini (`@google/genai`) is the runtime AI feature. Claude Code (Anthropic), including its subagents on 2026-10-20/21, generated most code drafts, the docs and the synthetic data. Google AI Studio was used for the key and manual prompt trials. Everything was reviewed by the team and verified by tests, CI, evals, an end-to-end API run and screenshots. Every member walks through code they didn't write (CHECKLIST P11.44–P11.49).

11. Limitations

Synthetic data. Rigid box-shaped products only (no clothing). One carton per order. Illustrative costs. No live Salla/Zid integration. Time-bounded heuristic (not globally optimal, although all 20/20 exact-fill fixtures are found). No crush/stacking-load or void-fill modeling. The 70% support and 1.5 cm padding are assumptions.

12. Next steps

Pilot with one merchant (100 real orders, packer vs Rassah: carton spend, courier invoices, damage, packing time). Salla/Zid integration. Multi-carton orders. Manufacturer-spec lookup. Scale-weight feedback to catch bad facts. Separate preview/production databases.

13. References

- Project 1 brief (Tuwaiq Academy × Startupbootcamp × Meta), October 2026 (no URL in the repository).
- Gemini API docs: https://ai.google.dev/gemini-api/docs
- Gemini API pricing (as of 2026-10-21): https://ai.google.dev/gemini-api/docs/pricing
- Supabase docs: https://supabase.com/docs
- Vercel docs: https://vercel.com/docs
- Next.js docs: https://nextjs.org/docs (the version used is also bundled in `node_modules/next/dist/docs/`)
- React Three Fiber: https://r3f.docs.pmnd.rs
- Paccurate API documentation (existing cartonization product, used for positioning). No URL is recorded in the repository.
- py3dbp (`enzoruiz/3dbinpacking`, MIT): reference for the naive first-fit baseline idea. Our code is independent TypeScript. No URL is recorded in the repository.
- Courier volumetric-weight divisor 5000 cm³/kg: described in `docs/PLAN.md` as "commonly 5000" and configurable (`VOLUMETRIC_DIVISOR`). No courier page has been cited yet.
- Bin-packing papers: none read or cited.